HW4 12209009S 杜奕旋

P1 $\tilde{Y}_1 = (1+r_f)(Y_0 - a) + (1+\tilde{r})a = Y_0(1+r_f) + a(\tilde{r} - r_f)$

$\max E[u(\tilde{Y}_1)] = \max\{u[Y_0(1+r_f) + a(\tilde{r} - r_f)]\}$

$E(\tilde{r}) = \frac{1}{2} \times 0.05 + \frac{1}{2} \times 0.3 = 0.175 > r_f \quad \therefore a^* > 0$

First-order condition: $E\{u'[Y_0(1+r_f) + a^*(\tilde{r} - r_f)](\tilde{r} - r_f)\} = 0$

$E\left[\frac{\tilde{r} - r_f}{u'[Y_0(1+r_f) + a^*(\tilde{r} - r_f)]}\right] = 0 \quad \therefore a^* = 825$

P2 $\tilde{R} = \begin{cases} 1+r_A, & \pi \\ 1+r_B, & 1-\pi \end{cases}$

$\Leftrightarrow \max u(Y_0 - s) + \beta E[u(s\tilde{R})]$

First-order condition: $-u'(Y_0 - s) + \beta E[u'(s\tilde{R})\tilde{R}] = 0$

$\therefore \frac{1}{s - Y_0} + \beta\pi\frac{1+r_A}{s(1+r_A)} + \beta(1-\pi)\frac{1+r_B}{s(1+r_B)} = 0$

$\therefore s^* = \frac{\beta Y_0}{1+\beta}$ $\therefore$ The optimal saving doesn't depend on π.

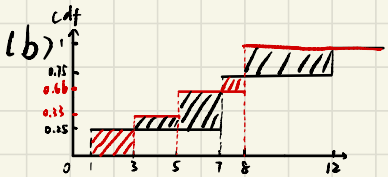
And the result cannot be generalized

P3 (a) $E(\tilde{z}_1) = 6.75 \quad E(\tilde{z}_2) = 5.36 \quad E(\tilde{z}_1) > E(\tilde{z}_2)$

$Var(\tilde{z}_1) = 15.1875 \quad Var(\tilde{z}_2) = 4.25 \quad Var(\tilde{z}_1) > Var(\tilde{z}_2)$

$\therefore$ No dominance relationship.

(b)



No dominance relationship

(c) $\int_0^3 (F_2 - F_1)\,dx = 0.5 > 0$

$\int_0^5 (F_2 - F_1)\,dx = 0.34 > 0$

$\int_0^7 (F_2 - F_1)\,dx = -0.48 < 0$

$\therefore$ No dominance relationship